

6

Chemical Equations

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6.1 Writing Chemical Equations

A chemical equation is a shorthand way of describing a chemical reaction. It uses symbols of elements and formulas of compounds in place of words to describe a chemical change or reaction.

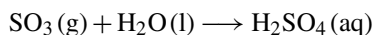
6.1.1 General rules for writing chemical equations

1. The formulas and symbols of reactants are written on the left side of the equation.
2. The formulas and symbols of products are written on the right side of the equation.
3. A plus sign (+) is placed between different reactants and different products.
4. The reactants are separated from the products by an arrow ($\longrightarrow$) pointing in the direction of the reaction. For a reversible reaction, a double arrow ($\rightleftharpoons$) is used.
5. The physical states of substances may be indicated by the symbols (s) for solid, (l) for liquid, (aq) for aqueous solution, and (g) for gases.
6. The equation is then balanced by inserting appropriate coefficients for each product and reactant.

Consider the following word equation:

Gaseous sulfur trioxide reacts with liquid water to form aqueous sulfuric acid.

We can substitute the formulas for reactants and products, following the above rules, and obtain a chemical equation for the reaction as



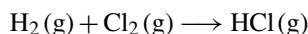
6.2 Balancing Chemical Equations

An equation as translated from words may not be balanced. It only shows the substances present—reactants and products. A chemical equation is balanced when it has the same number of atoms of each element on either side of the equation. Thus, a balanced equation obeys the law of conservation of mass. That is, atoms are not “created” or “destroyed” in writing a chemical equation.

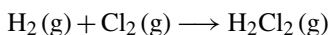
6.2.1 Guidelines for balancing a chemical equation

Balance the equation to make sure the law of conservation of mass is observed.

1. Inspect both sides of the equation to identify atoms that need to be balanced.
2. Balance one element at a time by placing a suitable coefficient to the left of the formula containing the element. Note that a coefficient placed in front of the formula affects all the atoms in the formula. For example, 2 H₂O implies two molecules of water containing four atoms of hydrogen and two atoms of oxygen.
3. Never attempt to balance an equation by changing subscripts because this will change the formulas of the compounds. For example, to balance the equation

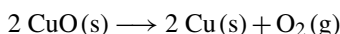


you cannot change the subscripts as below



because H₂Cl₂ is *not* the same as the compound HCl.

4. Make sure the elements already balanced are not changed since we are essentially doing this one element at a time.
5. Balance polyatomic ions (e.g. NH₄⁺ and CO₃²⁻) as single entities if they appear unchanged on both sides of the equation.
6. The balanced equation should contain the smallest possible set of whole-number coefficients. For example, a balanced equation showing the decomposition of copper (II) oxide should be expressed as:



And not as:

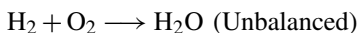


7. Do a final check by counting atoms on each side of the equation.

You may find some equations difficult to balance using these rules, for example, redox equations. These will be treated under Oxidation and Reduction Reactions, in Chapter 22.

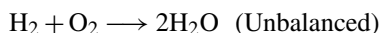
Example 6.1

Balance the equation

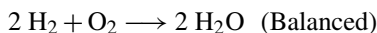


Solution

Hydrogen is balanced since there are two atoms of hydrogen on each side of the equation. Oxygen is not balanced. There are two atoms on the left hand and one atom on the right side. To balance oxygen, place the coefficient 2 in front of H₂O.



Oxygen is now balanced but hydrogen is no longer balanced. To balance hydrogen, place the coefficient 2 in front of H_2 .



There are four atoms of hydrogen and two atoms of oxygen on either side of the equation. The equation is now balanced.

Example 6.2

Sodium metal reacts with water to form sodium hydroxide and hydrogen. Write a balanced chemical equation for this reaction.

Solution

Sodium + Water $\longrightarrow$ Sodium hydroxide + hydrogen



By inspection, Na and O are balanced on both sides of the equation, but hydrogen is not. Balance hydrogen by placing the coefficient 2 in front of H_2O and NaOH.



All the atoms except Na are balanced. Balance Na by placing the coefficient 2 in front of it.



Final check: There are four atoms of H, two atoms of O, and two atoms of Na on either side of the equation. The equation is balanced.

Example 6.3

Balance the following chemical equations:

- (a) $\text{Fe} + \text{O}_2 \longrightarrow \text{Fe}_2\text{O}_3$
- (b) $\text{Fe}_2\text{O}_3 + \text{CO} \longrightarrow \text{Fe} + \text{CO}_2$
- (c) $\text{H}_3\text{PO}_4 + \text{Ca}(\text{OH})_2 \longrightarrow \text{Ca}_3(\text{PO}_4)_2 + \text{H}_2\text{O}$
- (d) $\text{C}_2\text{H}_3\text{Cl} + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{HCl}$

Solution

- (a) $4 \text{Fe} + 3 \text{O}_2 \longrightarrow 2 \text{Fe}_2\text{O}_3 \text{ (Balanced)}$
- (b) $\text{Fe}_2\text{O}_3 + 3 \text{CO} \longrightarrow 2 \text{Fe} + 3 \text{CO}_2 \text{ (Balanced)}$
- (c) $2 \text{H}_3\text{PO}_4 + 3 \text{Ca}(\text{OH})_2 \longrightarrow \text{Ca}_3(\text{PO}_4)_2 + 6 \text{H}_2\text{O} \text{ (Balanced)}$
- (d) $2 \text{C}_2\text{H}_3\text{Cl} + 5 \text{O}_2 \longrightarrow 4 \text{CO}_2 + 2 \text{H}_2\text{O} + 2\text{HCl} \text{ (Balanced)}$

6.3 Types of Chemical Reactions

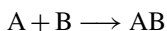
There are numerous types of chemical reactions. Most conform to one of the following:

- Combination or synthesis
- Decomposition
- Displacement
- Double decomposition or metathesis
- Neutralization
- Oxidation-reduction.

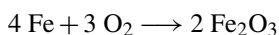
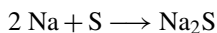
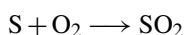
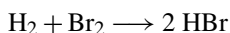
The above classification is helpful when writing chemical equations and can assist in predicting the products of other reactions.

6.3.1 Combination or synthesis

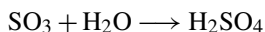
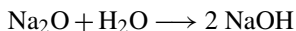
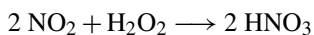
A combination reaction is one in which two or more reactants (elements or compounds) combine to form a single product. The general equation for a combination is:



Some examples involving elements include:

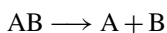


Some examples involving compounds are:

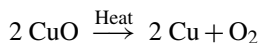
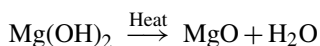
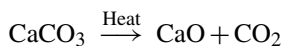
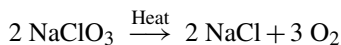


6.3.2 Decomposition

A decomposition reaction is one in which a single reactant is broken down into two or more simpler substances, usually under the action of heat. The general equation for a decomposition reaction is

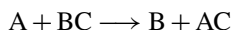


Representative examples of decomposition reactions include:

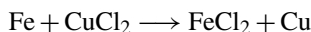
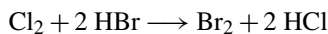
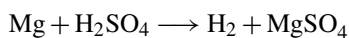


6.3.3 Displacement

A displacement reaction is one in which a free element replaces another element within a compound. The general equation for a displacement replacement is

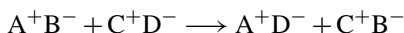


Representative examples involving metals and non-metals include:

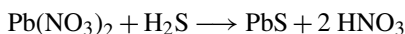
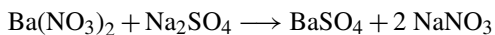
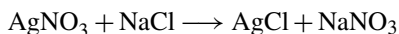


6.3.4 Double decomposition or metathesis

A double decomposition (or metathesis) reaction is one in which two compounds exchange ions with each other to form two new compounds. The general equation for a double decomposition reaction is

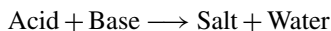


Examples include:

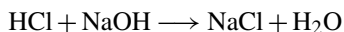
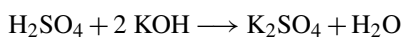


6.3.5 Neutralization

A neutralization reaction is one in which an acid reacts with a base to form a salt and water only. The general expression is:



Some examples include:



This is a special form of double displacement reaction. Acid-base reactions are discussed in Chapter 14 and Chapter 18.